A sharp lower general-position bound for Cartesian products

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Abstract

The lower general-position number of a finite graph is the smallest cardinality of an inclusion-maximal vertex set containing no three vertices on a geodesic. We refute the conjecture that the lower general-position number of a Cartesian product is at least the smaller of the two factor parameters. For every integer $r \geq 6$, we construct connected diameter-two factors of orders $2r-1$ and $4r+5$ whose smaller factor parameter is r, while their product has lower general-position number exactly five. Thus the conjectured bound can fail by an arbitrarily large additive amount and an arbitrarily small ratio. We prove the sharp replacement $\mathrm{gp}^-(G\square H) \geq \min\{5, \mathrm{gp}^-(G), \mathrm{gp}^-(H)\}$ for all nonempty finite simple graphs. The lower bound has a computer-assisted proof reducing arbitrary distances to equality patterns in the five-point pseudometric cone, followed by an exact finite obstruction check. The construction and unbounded-gap argument are elementary. An accompanying Lean proof verifies the fixed 11-by-29 counterexample at the level of actual shortest paths; it does not formalize the universal bound.

Keywords: lower general position; Cartesian product; graph distance; counterexample; metric cone; computer-assisted proof.
MSC 2020: 05C12, 05C69, 05C76.

1 Introduction

For a finite graph G, a general-position set contains no three distinct vertices on any shortest path. Its lower general-position number $\mathrm{gp}^-(G)$ minimizes cardinality over *inclusion-maximal* general-position sets. Unlike the ordinary general-position number, it measures the smallest terminal outcome of adding vertices while maintaining general position. The parameter was studied systematically by Di Stefano, Klavžar, Krishnakumar, Tuite and Yero [2]. Products expose a particular difficulty: maximality in a factor need not translate into maximality in a product.

Kruft Welton, Khudairi and Tuite [1] proposed that

$$\mathrm{gp}^-(G\square H) \geq \min\{\mathrm{gp}^-(G), \mathrm{gp}^-(H)\}. \tag{1}$$

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This is Conjecture 3 on page 117 of their published paper and Conjecture 2.10 in arXiv:2404.19451v1. We give connected counterexamples and determine the best universal lower bound depending only on the smaller factor parameter.

Theorem 1.1. *For all nonempty finite simple undirected graphs G, H,*

$$\mathrm{gp}^-(G\Box H) \geq \min\{5, \mathrm{gp}^-(G), \mathrm{gp}^-(H)\}. \tag{2}$$

Theorem 1.2. *For every integer $r \geq 6$, there are connected diameter-two graphs G_r, H_r with $|V(G_r)| = 2r-1$ and $|V(H_r)| = 4r+5$ such that*

$$\mathrm{gp}^-(G_r) = r, \qquad \mathrm{gp}^-(H_r) \geq r, \qquad \mathrm{gp}^-(G_r\Box H_r) = 5.$$

The proof of the first two assertions and of the product upper bound 5 is elementary and does not use Theorem 1.1.

Corollary 1.3. *For each positive integer r, define*

$$\Phi(r) = \min\{\mathrm{gp}^-(G\Box H) : G, H \text{ are nonempty finite simple graphs, } \min\{\mathrm{gp}^-(G), \mathrm{gp}^-(H)\} = r\}.$$

Then $\Phi(r) = \min\{r, 5\}$. In particular, (1) holds whenever one factor has lower general-position number at most five, but no unbounded function of the smaller factor parameter is a universal product lower bound.

The five-point set in our construction uses three first-coordinate roles and five second-coordinate roles. Large true-twin cliques increase the factor parameters without changing the distance relations that block extensions in the product. This separates factor saturation from product saturation and explains the unbounded failure of (1).

The universal statement requires a different argument. A maximal product set with fewer than five vertices would impose a finite collection of incompatible factor-extension demands. Arbitrary factor distances are reduced to the zero patterns of distances and triangle slacks in MET_5. The finite obstruction check uses integer arithmetic throughout. Its derivation, exhaustive-search invariant and output are given in Sections 4–5.

Relation to prior work

The complete-factor formula is already known [1]. That paper's unbounded-gap examples concern factor lower *terminal* numbers, rather than the factor lower general-position numbers used here. We include the familiar complete-factor equality cases for sharpness.

True-twin saturation and clique enlargement are established tools; related constructions appear in Brešar and Yero [3, Section 2.2, Proposition 5]. We prove the required saturation lemma explicitly and do not claim that general mechanism as new. The halfspace-clipping method used below is standard polyhedral theory [4]. Our contribution is the particular counterexample family, the sharp product bound and the finite profile obstruction establishing that bound. Literature checks did not locate a verified earlier solution with this scope; they do not certify historical priority.

2 Definitions and preliminary facts

All graphs in this paper are finite, simple and undirected. A graph is *nonempty* when it has at least one vertex. A geodesic is a shortest path between its endpoints. In a connected graph, distinct vertices x, y, z are collinear if one of

$$d(x,y) + d(y,z) = d(x,z), \quad d(x,z) + d(z,y) = d(x,y), \quad d(y,x) + d(x,z) = d(y,z)$$

holds. A subset is in general position, or is a GP set, if it has no collinear triple. This is equivalent to the shortest-path definition: a subpath of a geodesic is geodesic, and a distance equality concatenates two geodesics to a geodesic through the middle vertex. In disconnected graphs we use the shortest-path definition, so vertices in different components cannot occur on one path.

Definition 2.1. A GP set $S \subseteq V(G)$ is *maximal* if it is inclusion-maximal: $S \cup \{x\}$ fails to be GP for every $x \in V(G) \setminus S$. We set

$$\mathrm{gp}^-(G) = \min\{|S| : S \subseteq V(G) \text{ is a maximal GP set}\}.$$

The minimum exists by finiteness. GP is hereditary, so failure of all one-vertex extensions is equivalent to inclusion-maximality. Since G is nonempty, the empty set is never maximal and $\mathrm{gp}^-(G) \geq 1$.

The Cartesian product $G\Box H$ has vertex set $V(G) \times V(H)$. Two distinct vertices $(a, b), (a', b')$ are adjacent if either $a = a'$ and $bb' \in E(H)$, or $b = b'$ and $aa' \in E(G)$.

Lemma 2.2. *For connected G, H,*

$$d_{G\Box H}((a, b), (a', b')) = d_G(a, a') + d_H(b, b').$$

Proof. Every product path has at least $d_G(a, a')$ first-coordinate edge steps and at least $d_H(b, b')$ second-coordinate edge steps. Conversely, concatenate a first-coordinate geodesic with a second-coordinate geodesic. □

Lemma 2.3 (True-twin saturation)**.** *Let G be connected, and let x, y be distinct vertices with $N[x] = N[y]$. If a GP set contains x and omits y, adjoining y preserves general position. Consequently, every maximal GP set meeting a true-twin clique contains that entire clique.*

Proof. The equality of closed neighborhoods implies $xy \in E(G)$. Interchanging x, y and fixing all other vertices is an automorphism, so $d(x, w) = d(y, w)$ for $w \notin \{x, y\}$. A triple containing y but not x has the same distances as the corresponding triple with x. The distances of a triple x, y, w are $1, t, t$ with $t \geq 1$, and none satisfies a triangle equality. These are all new triples. Repeatedly apply this observation in a true-twin clique. □

Lemma 2.4. *In a connected graph of diameter at most two, a set is GP if and only if its induced subgraph is a disjoint union of cliques.*

Proof. A collinear triple must have distances $1, 1, 2$, so it is exactly an induced three-vertex path. A graph contains no induced three-vertex path if and only if each of its connected components is complete: a shortest path between nonadjacent vertices in a noncomplete component supplies such a triple. □

3 An elementary counterexample family

Fix $r \geq 6$. Let A, B be disjoint cliques of order $r - 1$ and adjoin a vertex c adjacent to every vertex of $A \cup B$, with no edges between A and B. Thus G_r consists of two copies of K_r intersecting only in c. Choose $a \in A$ and $b \in B$.

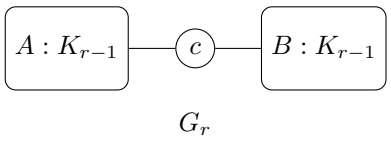
For H_r, begin with

$$L = \{z, u_0, u_1, v_0, v_1\}.$$

The subgraph induced by L is K_5 with u_0u_1 and v_0v_1 deleted. For each $i, j \in \{0, 1\}$, add a clique Q_{ij} of order r; its vertices are adjacent outside Q_{ij} precisely to z, u_i, v_j. There are no edges between

different Q-cliques. Equivalently, $\{z, u_i, v_j\} \cup Q_{ij}$ is a complete graph for each i, j. Both c and z are universal in their respective graphs. Nonadjacent vertices exist in each factor, so both graphs have diameter two. Their orders and edge counts are

$$\begin{aligned} |V(G_r)| &= 2r - 1, \qquad |E(G_r)| = r(r-1), \\ |V(H_r)| &= 4r + 5, \qquad |E(H_r)| = 2r^2 + 10r + 8. \end{aligned} \tag{3}$$



G_r

$$H_r = \bigcup_{i,j \in \{0,1\}} K_r[Q_{ij}] \vee K_3[\{z, u_i, v_j\}]$$

Four complete blocks, sharing only their indicated landmarks

Figure 1: Construction scheme. The two lines on the left indicate complete adjacency from c to the displayed cliques. On the right the union is a union of graphs on the stated shared vertex sets; each individual joined block is K_{r+3}. No edges are added between different Q_{ij}.

Proposition 3.1. *The maximal GP sets of G_r are $\{c\} \cup A$, $\{c\} \cup B$ and $A \cup B$. Hence $\mathrm{gp}^-(G_r) = r$. Moreover, $\mathrm{gp}^-(H_r) \geq r$.*

Proof. A GP set of G_r containing c cannot meet both A and B, since $a' - c - b'$ is geodesic whenever $a' \in A, b' \in B$. It lies in one of the two K_r blocks, and maximality then requires the entire block. A GP set omitting c and contained in one side can add c; it is not maximal. The set $A \cup B$ is GP by Lemma 2.4, and any of its proper subsets can add a missing vertex from that set. These cases give the stated classification and minimum r.

Each Q_{ij} is a true-twin clique. Lemma 2.3 implies that any maximal GP set meeting it has cardinality at least r. Suppose instead that a maximal GP set is contained in L. It must be maximal within L as well. The maximal GP subsets of L are exactly

$$\{u_0, u_1\}, \quad \{v_0, v_1\}, \quad \{z, u_i, v_j\} \quad (i, j \in \{0, 1\}). \tag{4}$$

Indeed, a GP set containing z must be a clique. A nonclique GP subset of L contains one of the two nonadjacent pairs, and any additional landmark creates an induced three-vertex path. Any smaller clique can be enlarged to one of the displayed triangles.

A vertex of Q_{ij} extends $\{z, u_i, v_j\}$ to a clique. A vertex of Q_{00} extends either nonadjacent pair in (4), since it is adjacent to exactly one vertex of that pair. Thus every displayed set has an extension outside L, a contradiction. No maximal GP set is contained in L, proving $\mathrm{gp}^-(H_r) \geq r$. □

Proposition 3.2. *The set*

$$S = \{s_0, s_1, s_2, s_3, s_4\} = \{(a, u_0), (a, u_1), (c, z), (b, v_0), (b, v_1)\} \tag{5}$$

is a maximal GP set of $G_r \square H_r$. Consequently $\mathrm{gp}^-(G_r \square H_r) \leq 5$.

Proof. Lemma 2.2 gives the selected distance matrix

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 3 \\ 2 & 0 & 2 & 3 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 3 & 2 & 2 & 0 \end{pmatrix}. \tag{6}$$

The sum of two nonzero entries is at least four, larger than every selected pair distance. Hence S is GP.

Write $x = (g, h) \notin S$. Every $h \neq z$ is adjacent to z and has distance two from at least one of u_0, u_1, v_0, v_1: a landmark is nonadjacent to its paired landmark, while a vertex of Q_{ij} is nonadjacent to u_{1-i}, v_{1-j}.

If $g = c$, then $h \neq z$. Choose a selected point s_t whose second coordinate has distance two from h. We have

$$d(x, s_2) = 1, \qquad d(s_2, s_t) = 2, \qquad d(x, s_t) = 3.$$

Thus s_2 lies on an x–s_t geodesic.

Suppose $g \in A$. If $h = z$, then $d(x, s_2) = 1$, $d(s_2, s_3) = 2$ and $d(x, s_3) = 3$. If $h \notin \{z, u_0, u_1\}$, one of v_0, v_1 has distance two from h: this is immediate for either v-landmark and for a Q-vertex. With $t = 3 + j$ for such a v_j, we obtain

$$d(x, s_2) = 2, \qquad d(s_2, s_t) = 2, \qquad d(x, s_t) = 4.$$

Finally, if $h = u_i$, the fact that $x \notin S$ implies $g \neq a$, and

$$d(x, s_i) = 1, \qquad d(s_i, s_{1-i}) = 2, \qquad d(x, s_{1-i}) = 3.$$

Each subcase creates a collinear triple with two selected vertices.

For $g \in B$, apply the same argument under the graph symmetry exchanging A, B, a, b, u_i, v_i and Q_{ij}, Q_{ji}. The first-coordinate roles c, A, B exhaust $V(G_r)$, so every outside vertex is blocked. Thus S is maximal. □

Corollary 3.3. *The conjecture* (1) *is false even for connected factors of diameter two. For the family above its additive deficit is at least $r - 5$, and the ratio of product lower general-position number to the smaller factor parameter is at most $5/r$.*

Proof. Combine Propositions 3.1 and 3.2. □

4 Reducing the universal bound to metric equality patterns

We first assume that G, H are connected. If Theorem 1.1 fails, there is a maximal GP set

$$S = \{(a_i, b_i) : 0 \leq i < k\}, \qquad 1 \leq k \leq 4, \quad k < \mathrm{gp}^-(G), \mathrm{gp}^-(H).$$

The selected product vertices are distinct, but their coordinate projections can repeat. The arrays

$$A_{ij} = d_G(a_i, a_j), \qquad B_{ij} = d_H(b_i, b_j)$$

are therefore *pseudometrics* on the k labels: they are symmetric, nonnegative and zero on the diagonal, and satisfy the triangle inequalities. Zero distance identifies repeated actual vertices.

For three labels, a triangle equality holds in $A + B$ if and only if the corresponding equality with the *same middle label* holds in both A and B. This follows because the two triangle slacks are nonnegative. Likewise two product labels coincide if and only if their distance is zero in both coordinates.

4.1 Profiles and compatibility

For a factor vertex x and an unordered landmark pair $i < j$, record three bits, in the order

$$\begin{aligned} d(a_i, x) + d(x, a_j) &= d(a_i, a_j), \\ d(x, a_i) + d(a_i, a_j) &= d(x, a_j), \\ d(x, a_j) + d(a_j, a_i) &= d(x, a_i). \end{aligned} \tag{7}$$

Let $E(x)$ be the set of the $3\binom{k}{2}$ positions whose bits are true, and let $Z(x) = \{i : d(x, a_i) = 0\}$. We call $(E(x), Z(x))$ its profile. Use the analogous definition for the second factor. For an outside factor vertex $x \notin \{a_0, \ldots, a_{k-1}\}$, $Z(x) = \varnothing$. Landmark vertices have their mandatory known profiles. Multiple bits in (7) may be true when landmark coordinates repeat, and are never replaced by a single mutually exclusive case.

Lemma 4.1. *If S is GP, a factor-profile pair $(E, Z), (E', Z')$ corresponds to a one-vertex GP extension of S precisely when*

$$Z \cap Z' = \varnothing, \qquad E \cap E' = \varnothing. \tag{8}$$

Here the pair must be realized by actual factor vertices. We call any pair of abstract profiles satisfying (8) compatible.

Proof. The first condition says that the product point is not one of the selected points. Since S is GP, a new collinear triple must consist of that point and two selected points. By distance additivity and nonnegative slacks, it occurs exactly when both factors have an equality in the same one of the three positions in (7). This is excluded by the second condition. □

4.2 A complete finite cover of all distances

Let $C = \mathrm{MET}_5 \subseteq \mathbb{R}^{10}$ be the cone of all pseudometrics on $\{0, 1, 2, 3, 4\}$, with coordinates d_{ij} for $0 \le i < j \le 4$. Its defining inequalities are the ten coordinate nonnegativities and the thirty triangle inequalities

$$d_{ij} \ge 0, \qquad d_{ij} + d_{j\ell} - d_{i\ell} \ge 0 \quad (i, j, \ell \text{ distinct}). \tag{9}$$

Each of the three choices of middle label is used once for each unordered triple. A pseudometric on the k landmarks together with a new vertex extends to C: put the new vertex at label 4 and fill unused labels by copies of label 0. Thus this cone covers every distance pattern needed for $k \le 4$, without bounding graph order, diameter or distances.

Lemma 4.2 (Exact ray generation)**.** *The halfspace-clipping algorithm below enumerates all extreme rays of C, using integer arithmetic. Its output has 25 rays.*

Proof. Begin with the ten unit rays of the nonnegative orthant and add the triangle inequalities in lexicographic triple order. If the next inequality is $h \ge 0$, retain every old ray r with $h(r) \ge 0$. For each adjacent old ray pair r, s with $h(r) > 0 > h(s)$, add

$$h(r)s - h(s)r, \tag{10}$$

normalized by the positive greatest common divisor of its coordinates. The active inequalities of each ray are kept as a bit mask. Two distinct old rays are adjacent exactly when no third old ray vanishes on all their common active inequalities.

For completeness, every intermediate cone is pointed, since it lies in the nonnegative orthant. It is full-dimensional, since the all-one vector satisfies each coordinate and triangle inequality strictly.

Intersecting with $\sum_{i<j} d_{ij} = 1$ gives a compact polytope. Clipping by a halfspace retains the feasible old vertices and creates new vertices only on old edges that cross its boundary. The minimal face containing two old vertices is defined by their common active inequalities; it is an edge if and only if it contains no third vertex. These observations justify both the adjacency test and update (10) inductively, starting from the orthant. Every operation is exact rational, and the displayed integer normalization avoids denominators. Running this update for the thirty explicitly specified inequalities gives 25 rays; the independent replay derives them from (9), rather than taking a ray table as input. Its intermediate counts are given in Appendix A. □

Let $\mathcal{F}$ be the forty nonnegative linear forms in (9). For $d \in C$, set

$$\mathcal{Z}(d) = \{f \in \mathcal{F} : f(d) = 0\}.$$

Lemma 4.3 (Zero-pattern closure)**.** *All zero patterns $\mathcal{Z}(d)$ in C are the intersections of subsets of the* 25 *extreme-ray zero patterns. The empty intersection is $\mathcal{F}$, the pattern of the origin. There are* 9484 *distinct patterns. Each has an integer pseudometric representative.*

Proof. Write $d = \sum_{r \in R} \lambda_r r$ for the extreme rays R and $\lambda_r \geq 0$. Since each $f \in \mathcal{F}$ is nonnegative on every ray, $f(d) = 0$ if and only if $f(r) = 0$ for every ray with $\lambda_r > 0$. Hence

$$\mathcal{Z}(d) = \bigcap_{\lambda_r > 0} \mathcal{Z}(r).$$

Conversely, the sum of the rays in any chosen subset realizes that intersection; the empty subset gives the origin. The checker starts with $\mathcal{F}$ and repeatedly intersects each reached mask with each ray mask. A newly reached mask is stored with the sum of the corresponding ray and the preceding representative, whose forty zero tests are checked exactly. Queue exhaustion gives the entire intersection closure, yielding 9484 patterns. This enumerates *equality types*, not metrics up to a distance cutoff. □

4.3 Landmark types and extension demands

The *landmark type* of a pseudometric records which landmark distances are zero and which landmark triangle slacks are zero. Group the patterns of Lemma 4.3 by their landmark type, retaining patterns for which $d_{i4} > 0$ for every $0 \leq i < k$. Every such pattern contributes an outside profile. Every actual outside factor vertex is included, and all known profiles are determined by the landmark type.

This step allows profiles having the same equality type but different numerical landmark distances. It is a relaxation: the collected profiles need not have a simultaneous realization in one graph. Nonexistence for the relaxed system suffices, since it includes every actual factor system.

For a fixed landmark type, identify zero-distance labels and choose one representative per distinct landmark. Let T range over all maximal GP subsets of these distinct landmarks. For each T define its demand

$$\mathcal{E}_T = \{E : \ E \text{ is an outside profile and no bit for a pair in } T \text{ belongs to } E\}.$$

The phrase "a bit for a pair in T" includes all three positions in (7) for that pair.

Lemma 4.4. *If a factor has lower general-position number larger than k, its set X of actual outside profiles intersects every $\mathcal{E}_T$.*

Proof. The set T has at most k vertices, so it is not maximal in the entire factor. It has an extending vertex. No distinct landmark can extend it, by the choice of T. Thus the extending vertex is outside the landmark set, and its profile has none of the prohibited pair bits. □

If S were maximal, Lemma 4.1 would forbid compatible pairs among all known and actual outside profiles. At the same time the outside profiles on both sides would satisfy every demand of Lemma 4.4. The remainder of the connected-factor proof excludes this finite system.

5 Exact obstruction certificate and completion of the bound

Proposition 5.1 (Finite obstruction)**.** *For each $k \in \{1, 2, 3, 4\}$, no pair of landmark types can describe a GP product set of size k with outside-profile sets satisfying all demands and avoiding every compatible pair, including pairs involving known profiles.*

Computer-assisted proof. The data are generated by Lemmas 4.2 and 4.3. For every unordered pair of landmark types, reject pairs having a common zero landmark distance or a common landmark triangle equality: they do not describe distinct product vertices in general position. Factor exchange makes unordered pairs sufficient.

If two known profiles are compatible, they already give a product extension. Otherwise, let A_G, A_H be the outside profiles remaining after deleting all those compatible with a known profile on the opposite side. Intersect each demand with its remaining profiles.

An exhaustive search selects second-factor profiles. Its state is (U, A), where U is the set of unmet second-factor demands and A is the set of still permissible first-factor profiles. Initially every second demand is unmet and $A = A_G$. If some first demand has empty intersection with A, the state is impossible. Otherwise choose an unmet second demand and branch over *every* profile q remaining in it. Remove from U all demands hit by q, and remove from A all profiles compatible with q. Repeated states may be merged.

If U becomes empty while every first demand still meets A, choosing all profiles in A gives a feasible relaxed system. Indeed, profiles within one factor have no additional compatibility requirement; only cross-factor compatibility is relevant. The checker finds no such state.

This search is complete. Any hypothetical actual second-factor set must contain a profile in the selected unmet demand and follows one of the enumerated branches. All deleted first profiles are forbidden by that selection. Each step meets a new second demand, so depth is finite. Merging identical (U, A) states is valid because future feasibility depends only on these two objects. An empty demand has no branch and is correctly rejected.

The exact output is shown in Table 1. The replay scripts generate the cone, zero patterns, demands and search from their definitions; no discovery table or solver output is read as proof input. All four sizes return infeasibility for every surviving type pair. □

Proof of Theorem 1.1. For connected factors, a counterexample would give a maximal product GP set of size $1 \le k \le 4$ with $k < \mathrm{gp}^-(G), \mathrm{gp}^-(H)$. Its actual factor profiles satisfy the necessary demands and have no compatible pair, contradicting Proposition 5.1.

For disconnected factors, write their nonempty connected components as G_i and H_j. GP and maximality are componentwise, because a path cannot connect different components. Therefore

$$\mathrm{gp}^-(G) = \sum_i \mathrm{gp}^-(G_i), \qquad \mathrm{gp}^-(H) = \sum_j \mathrm{gp}^-(H_j),$$

k	Landmark types	Outside profiles	GP type pairs	Search states
1	1	1	1	2
2	2	5	2	3
3	8	87	14	15
4	106	9101	2381	2508

Table 1: Exact obstruction replay. The profile column is summed over all landmark types. Known-profile obstructions can reject type pairs before a search state is entered. The generator has 25 rays and 9484 zero patterns in each run. The state totals include one additional state from the demand-removal negative control in each run.

and the components of the product are $G_i \square H_j$. Put $x_i = \mathrm{gp}^-(G_i)$ and $y_j = \mathrm{gp}^-(H_j)$. The connected result gives

$$\mathrm{gp}^-(G \square H) \geq \sum_{i,j} \min\{5, x_i, y_j\}.$$

For nonnegative t, y_j, the elementary inequality $\sum_j \min\{t, y_j\} \geq \min\{t, \sum_j y_j\}$ follows by considering whether some summand reaches t. Apply it first with $t = \min\{5, x_i\}$ and then sum over i, to obtain

$$\sum_{i,j} \min\{5, x_i, y_j\} \geq \sum_i \min\{5, x_i, \textstyle\sum_j y_j\} \geq \min\{5, \textstyle\sum_i x_i, \sum_j y_j\}.$$

This proves (2) for all stated graphs. □

Proof of Theorem 1.2 and Corollary 1.3. For $r \geq 6$, Theorem 1.1 supplies the lower bound five, while Proposition 3.2 supplies the matching upper bound. The factor parameters are given by Proposition 3.1.

For $1 \leq r \leq 5$, take $G = H = K_r$, both with lower GP number r. If a GP subset of $K_r \square K_r$ has cardinality less than r, an unused first coordinate and an unused second coordinate give a point at distance two from every selected point. It can be adjoined: selected pair distances are at most two, whereas a new triangle has distances $2, 2, t$ with $1 \leq t \leq 2$. A full row is GP and maximal when $r \geq 2$, since every outside point is adjacent to the row point in its column, which lies on a geodesic to any different row point. The case $r = 1$ is the singleton graph. Thus the product has lower GP number r. These equality examples and the universal bound show $\Phi(r) = \min\{r, 5\}$ for every positive integer r. □

6 Formal verification and reproducible computation

6.1 The fixed original-conjecture counterexample

Set $r = 6$. Label the vertices of G_6 by $c = 0$, $a_i = 1 + i$ and $b_i = 6 + i$ for $0 \leq i < 5$. Label H_6 by

$$u_0 = 0, \quad u_1 = 1, \quad z = 2, \quad v_0 = 3, \quad v_1 = 4, \quad Q_{ij} = \{5 + 6(2i + j), \ldots, 10 + 6(2i + j)\}.$$

The selected coordinate pairs are $(1, 0), (1, 1), (0, 2), (6, 3), (6, 4)$. The factor orders are $11, 29$, and their product has 319 vertices. Its edge count is

$$29 \cdot 30 + 11 \cdot 140 = 2410.$$

The accompanying file `LowerGPProduct319.lean` defines actual finite `SimpleGraph` factors and their Mathlib Cartesian product `boxProd`. Its GP definition quantifies over actual shortest simple

`Walk`s and their vertex positions. The proof establishes equivalence with the distance formulation, equivalence of one-point and inclusion maximality, and the genuine minimum cardinality among maximal GP sets. Its main fixed-object theorem proves

$$G_6, H_6, G_6 \square H_6 \text{ connected}, \qquad \mathrm{gp}^-(G_6) \geq 6, \quad \mathrm{gp}^-(H_6) \geq 6, \quad \mathrm{gp}^-(G_6 \square H_6) \leq 5.$$

It also refutes the universal conjecture on the fixed finite vertex types 11 and 29. These statements suffice to formalize the original counterexample, without assuming a supplied distance table or the universal-five theorem. The equality $\mathrm{gp}^-(G_6) = 6$ is proved above in ordinary mathematics; it is not a separate audited equality endpoint in this Lean file.

On 12 October 2026 the unchanged frozen source was compiled afresh with Lean 4.34.1 and Mathlib commit `d13f23b723b8a846827a245b89c10fc7d3f11612`, generating a new `.olean` object. All fifteen audited declarations used only `propext`, `Classical.choice` and `Quot.sound`; the successful run had no error or warning. Full logs, source hashes, toolchain identifiers and the fresh-object hash accompany the manuscript. The all-parameter family, the universal bound and product equality five are *not* covered by this fixed-object Lean proof.

6.2 Exact certificate replay

The supplement contains three principal replay commands, using only the Python standard library:

```
python supplement/fixed319/verify.py
python supplement/universal/verify_universal_five.py
python supplement/universal/verify_unbounded_gap.py
```

The first reconstructs adjacency independently, computes all factor and product BFS distances, compares factor BFS with Floyd–Warshall, and checks all 319^2 product distance entries against additivity. It verifies all ten selected triples and actual shortest blocking paths for all 314 outside vertices. Factor subsets of cardinality at most five are checked directly, along with the three maximal GP sets of G_6.

The second regenerates the metric rays, all equality masks and each of the four obstruction problems in Table 1. The third reconstructs the family at $r = 6, 7, 12, 31$ and checks the selected-distance and blocking mechanisms, as well as the complete-factor equality cases for $r = 1, \ldots, 5$. These finite regressions support the implementation; the proof for every $r \geq 6$ is Propositions 3.1 and 3.2, not extrapolation from those samples.

All three commands were executed successfully for the mathematical source version preserved from version 0.1; those execution records are retained. Negative controls reject false maximality, a false GP extension, an overstated factor lower bound and a genuine collinear factor triple. The universal checker also removes the factor-extension demands: the relaxed system then becomes feasible, as it should. It rejects a metric violating an original triangle inequality and recognizes a collinear three-point metric. These controls test meaningful failure modes; they are not additional proofs of exhaustiveness.

7 Discussion and scope

The obstruction occurs because maximality is not preserved under coordinate projection. In the family of Section 3, true-twin saturation forces large maximal GP sets in H_r, but the product can block every extension using only the five roles in (5). Increasing the clique sizes changes the factor parameters while leaving that blocking mechanism intact. The metric certificate explains why the

same phenomenon cannot produce a maximal product GP set of size at most four when both factor parameters are larger.

The sharpness is scalar: it determines the best bound from the smaller factor lower-GP number alone. It does not give an exact formula for every Cartesian product, classify all extremal pairs, minimize the order of a counterexample, or resolve the separate terminal-set existence conjecture in [1]. The construction establishes only the lower bound required for $\mathrm{gp}^{-}(H_r)$, without claiming its exact value for every r.

The universal proof has a finite computational component. Its mathematical reduction is independent of distance cutoffs, and its checker uses exact integer arithmetic with a complete search argument. Nevertheless, the entire universal proof is not Lean formalized. A purely noncomputational classification of the four-landmark obstruction would simplify this part of the argument; the elementary original-conjecture counterexample does not depend on such a classification.

AI assistance and responsibility

OpenAI Codex and a GPT-6-based agent system contributed substantially to mathematical exploration, construction and proof development, exact computation, Lean formalization, literature searches, independent AI checks, and manuscript writing. Separate AI reviewers examined the mathematical arguments and their correspondence to the original problem. Such review does not constitute external human peer review or certify historical originality. No AI system is listed as an author. River Zhang is the named human author; scientific responsibility for the final manuscript and any submission remains with the human author. No human-only research process is asserted.

A Replaying the finite obstruction

The halfspace order in `verify_metric_rays.py` takes unordered triples (a, b, c) lexicographically and, for each, the ordered triples $(a, b, c), (a, c, b), (b, a, c)$, interpreted as outer–middle–outer labels. Coordinates d_{ij} use lexicographic unordered-pair order. The numbers of extreme rays after the thirty successive triangle inequalities are

$$11, 11, 10, 12, 12, 11, 15, 15, 14, 19, 19, 20, 30, 30, 34,$$
$$49, 49, 64, 58, 52, 46, 42, 34, 30, 26, 22, 22, 23, 24, 25.$$

The complete ray-to-zero-pattern generator and obstruction checker are provided as executable source, without a third-party polyhedral or SAT dependency. The essential search can be written as follows; all sets are represented by bit masks in the implementation.

```
check_pair(G, H):
    if a known-known profile pair is compatible:
        return INFEASIBLE
    A = G.outside_profiles incompatible with every H.known
    B = H.outside_profiles incompatible with every G.known
    D = [demand intersect B for demand in H.demands]
    stack = [(all demand indices, A)]
    seen = empty set
    while stack is not empty:
        (U, A) = pop(stack)
        if (U, A) is in seen: continue
        add (U, A) to seen
        if some G.demand has empty intersection with A: continue
```

```
        if U is empty: return FEASIBLE
        choose an unmet demand D[i]
        for every q in D[i]:
            Unew = U minus all demand indices hit by q
            Anew = A minus every profile compatible with q
            push (Unew, Anew)
    return INFEASIBLE
```

Neither exhaustion nor a digest alone supplies the semantic connection to the original problem; that connection is proved by Lemmas 4.1 and 4.4. The exact code is bound to this manuscript by `source-bindings.json`. Newly executed outputs are in `verification/`; historical records in the source supplement retain their original dates.

The four deterministic search-stream SHA256 digests, for $k = 1, 2, 3, 4$, are:

```
96d806e0b6ceca36c5346273481079e390daa188fc97e4302f39b1dc0436133a
7772dbba1a8d4438b644d7b4fda05182389467038969c7629d5e9c239067d4e2
fa1fe74b732b9e524e0e13468067b8cb6651a50921f49632f025f4b5533c14ec
711f13a8995d9667bb276999ca636891a8ee82553193bbe068df0dc55ce1099f
```

These identify the replay streams under the fixed script version, not historical priority or independent mathematical truth. The supplement's manifest separately binds every included file by its SHA256 digest.

B Lean reproduction and provenance

The Lean source digest is

```
d1608a26b68eface444fc2c2feb53ebe8b24e364021213f589bd51e3aa52ab3e
```

Use Lean 4.34.1, the Mathlib revision specified in Section 6, and an existing cache or the ordinary Mathlib cache-download procedure. From a Lake project with those pinned dependencies, the source can be checked as

```
lake env lean supplement/fixed319/LowerGPProduct319.lean
```

When checking a source outside that project's directory while producing an object, pass `-R` with the source directory as the package root. The supplied fresh-object receipt records the actual local command and does not claim that an empty-cache machine was tested.

All mathematical inputs are bound to repository revision

```
61488e05c9fe41984f483853f99e79f5715dc893
```

of `https://github.com/mio-qwq/math`. The construction originates in the project's earlier frozen distributed result. Its construction and the universal certificate are retained, respectively, at

```
workstreams/ROOT/lower-gp-product319-counterexample/
workstreams/ROOT/distributed-written-proofs/B/lower_gp_product/
```

The manuscript gives the full mathematical argument rather than requiring the reader to reconstruct it from development logs. Exact source provenance and original version bindings accompany the supplement.

References

[1] E. Kruft Welton, S. Khudairi and J. Tuite, *Lower general position in Cartesian products*, Communications in Combinatorics and Optimization **10**(1) (2025), 110–125. doi:10.22049/cco.2024.29171.1879. Preprint: arXiv:2404.19451v1.

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[3] B. Brešar and I. G. Yero, *Lower (total) mutual-visibility number in graphs*, Applied Mathematics and Computation **465** (2024), 128411. doi:10.1016/j.amc.2023.128411. Preprint: arXiv:2307.02951v2.

[4] D. Avis, *Excursions in Polyhedral Computation using lrs*, author-hosted mathematical exposition, `https://cgm.cs.mcgill.ca/~avis/epc/epc.html`. Accessed 12 October 2026.